

Md Hasebul Hasan

Ph.D. Student in Computer Science and Engineering | Trustworthy, Human-Centered, and Efficient AI
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Research Interests

Human-AI interaction, natural language processing, trustworthy AI, LLM evaluation and safety, agentic systems, and efficient LLM inference. I develop methods and benchmarks that diagnose model failures, improve reliability in real-world interactions, and reduce the computational cost of deployment.

Education

Ph.D. in Computer Science and Engineering **Aug 2025 – Present**
University of Texas at Arlington, Arlington, TX *GPA: 4.00/4.00*
Advisor: Mohammad Atiqul Islam. Graduate Research Assistant.

M.Sc. in Computer Science and Engineering **Jul 2022 – Aug 2025**
Bangladesh University of Engineering and Technology (BUET), Dhaka *GPA: 3.75/4.00*
Thesis: *An Efficient Compositional Reasoning Approach for Solving Geospatial Queries Using LLMs*. Research led to publications at ICML, EACL, and IEEE COMPSAC.

B.Sc. in Computer Science and Engineering **Feb 2017 – May 2022**
Bangladesh University of Engineering and Technology (BUET), Dhaka *GPA: 3.77/4.00; Rank: 31/127*
Senior-year GPA: 3.89/4.00. Thesis: *A Multi-Modal Deep Learning Based Approach for House Price Prediction*.

Research Experience

Graduate Research Assistant **Aug 2025 – Present**
University of Texas at Arlington *Advisor: Mohammad Atiqul Islam*

- Research trustworthy, human-centered, and efficient LLM systems, with emphasis on real-world evaluation, small-language-model adaptation, and memory-efficient inference.
- Created **DeEscalWild**, a benchmark derived from real-world police-civilian interactions containing **1,500 scenarios, 285,887 dialogue turns, and approximately 4.7M tokens**; designed the transcription, diarization, filtering, anonymization, validation, and evaluation pipeline.
- Designed controlled evaluations across multiple 2B–3B language models, including fine-tuning, automatic metrics, LLM-as-a-judge evaluation, expert human review, and latency measurement.
- Investigate KV-cache compression for Llama-2-7B on NVIDIA H100 GPUs, studying low-rank projection, token retention, oracle allocation, K/V asymmetry, memory, throughput, and latency trade-offs.

Graduate Researcher – GeoAI and LLM Agents **2022 – 2025**
Bangladesh University of Engineering and Technology *M.Sc. research*

- Co-developed **MapEval**, a benchmark for geospatial reasoning in foundation models; led literature-gap analysis and helped design the spatial reasoning taxonomy, dataset construction, annotation, and human verification.
- Led **MapAgent**, a hierarchical agent with dynamic map-tool integration for compositional geospatial reasoning; first author on the EACL 2026 Findings paper.
- Developed **GuardFL**, a majority-consensus approach for detecting and removing malicious clients in federated learning; first author on the IEEE COMPSAC 2024 paper.

Publications

Peer-Reviewed / Accepted

1. **M. H. Hasan**, K. H. Charu, E. P. Sridhar, S. Deb, and M. A. Islam. “DeEscalWild: A Real-World Benchmark for Automated De-Escalation Training with SLMs.” *NeurIPS 2026 Evaluations and Datasets Track*, 2026. **Accepted as a Poster**. First author; led benchmark construction, data validation, model fine-tuning, and evaluation. [paper]
2. M. L. Dihan, M. T. Hassan, M. T. Parvez, **M. H. Hasan**, M. A. Alam, M. A. Cheema, M. E. Ali, and M. R. Parvez. “MapEval: A Map-Based Evaluation of Geo-Spatial Reasoning in Foundation Models.” *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. **Spotlight (top 2.6%)**. [paper]
3. **M. H. Hasan**, M. L. Dihan, T. Hashem, M. E. Ali, and M. R. Parvez. “MapAgent: A Hierarchical Agent for Geospatial Reasoning with Dynamic Map Tool Integration.” *Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026. [preprint]

4. **M. H. Hasan**, M. S. I. Khan, M. S. Haque, A. Y. Ahmed, and M. A. Adnan. “GuardFL: Detection and Removal of Malicious Clients Through Majority Consensus.” *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 1112–1117, 2024. [paper]

Manuscripts / Preprints

5. **M. H. Hasan**, M. A. Jahan, M. E. Ali, Y.-F. Li, and T. Sellis. “A Multi-Modal Deep Learning Based Approach for House Price Prediction.” arXiv:2409.05335, 2024. First author; led multimodal architecture design and evaluation. [preprint]

Industry Experience

Machine Learning Engineer
IQVIA, Dhaka, Bangladesh

May 2022 – Jul 2025
Full-time

- Built and deployed ML prediction systems using Python, PyTorch, TensorFlow, and Snowflake; production work contributed to solutions associated with a 15% increase in sales.
- Developed time-series data pipelines over enterprise data marts, including data engineering, preprocessing, model optimization, evaluation, and deployment.
- Architected LangChain-based multi-agent workflows, fine-tuned open-source LLMs for domain-specific applications, and conducted AI security red-team assessments.
- Implemented scalable MLOps with AWS SageMaker/Bedrock, Kubernetes, Kubeflow, Docker, ECR/EC2, Git, and GitLab CI/CD.

Teaching Experience

Lecturer, Department of CSE
Bangladesh University of Engineering and Technology (BUET)

Nov 2022 – Jul 2024
Part-time

- Taught six undergraduate sessional/laboratory courses: Algorithm Engineering, Software Development, Databases, Data Structures and Algorithms II, Digital Logic Design, and Structured Programming.
- Supervised **30 student projects**; designed and graded assignments, laboratory exercises, and examinations for classes of 40–60 students.

Research Grant

Research Grant for M.Sc. Thesis, Directorate of Advisory, Extension and Research Services, BUET, 2024. Competitive award of approximately **US\$1,500** supporting MapAgent research.

Awards & Honors

- Platinum Award, IQVIA Impact Program, 2024; Silver Award, IQVIA Impact Program, 2025.
- University Merit List Scholarship (two semesters).
- Dean’s List for three consecutive years (Levels 2–4), BUET.

Technical Skills

Programming: Python, C/C++, Java, JavaScript, SQL, LaTeX

ML / AI: PyTorch, TensorFlow, Hugging Face, scikit-learn, LangChain, LLM fine-tuning, RAG, agentic systems, model evaluation

Cloud / Systems: AWS SageMaker, Bedrock, EC2, ECR, Docker, Kubernetes, Kubeflow, Git/GitLab CI/CD, NVIDIA GPU/HPC workflows

Databases / Web: Snowflake, PostgreSQL, MySQL, MongoDB; Node.js, Vue.js, React

References

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